Note on the Existence of Perfect Maps*

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Summary—**In determining location in a previously mapped region by map-matching, there arises the question of minimum submap size relative to the size of the complete map of the region for unambiguous determination of position. A lower bound for the size of the submap is obtained for quantized binary maps. It is shown that there exist maps (called perfect) such that this lower bound is realized. Of special interest is the construction of a doubly periodic 4 × 4 perfect map for a 2 × 2 submap.**

The two-dimensional analogy of perfect maps to shift register codes suggests a possible development of planar error-correcting codes and an application to the two-dimensional range-velocity ambiguity problem of radar.

I. Introduction

THE general subject of this paper arises in the field of position location by map-matching. Let the map be approximated by a checker-board array of black and white squares. Represent this map by an array (or matrix if map is rectangular) of ones and zeros. Define the number of elements in the array to be the *area* of the map. For example, the matrix

$$M = \begin{bmatrix} 1 & 1 & 1 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 & 0 & 1 \\ 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 & 0 & 1 \end{bmatrix}$$

is a 6 × 5 map of area 30.

Now consider the 3 × 2 submap

$$M_s = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix}.$$

Observe that this submap matches two *positions* in the larger map above. Consequently, either the area of M_s is not sufficient or its shape is not proper to locate position uniquely in M. It is our purpose to investigate the relationship of size and shape of the submap to the uniqueness of position location by map-matching in the larger map.

Henceforth, let us restrict the discussion to rectangular maps so that every map is a matrix. For ease of discussion, further imagine that every element of this matrix occupies a parallelogram in a network of identical parallelograms which cover the surface of a torus. The upper and lower rows and left and right columns can be connected on the surface of the torus in a variety of ways. We demand here that the boundary identifications are such that a boundary element is connected to one and only one element and that a boundary element of one row (column) is connected to either a boundary element of the same or adjacent row (column).

Assume an $M \times N$ map on a torus. Consider an $m \times n$ parallelogram submap where $m \leq M$ and $n \leq N$. There are 2^{mn} possible submaps of the given shape and area. In order that each of these submaps uniquely specify a position on the torus, there must be at least as many submaps as there are positions (elements) in the larger map. Thus, for uniqueness of position location it is *necessary* that

$$2^{mn} \geq MN.$$

A *perfect* $M \times N$ map on a torus is an $M \times N$ matrix of ones and zeros such that each of the 2^{mn} submaps of area

$$mn = [\log_2 MN]$$

defines a unique position on the torus, where $[X]$ denotes least integer greater than X. In the next section we demonstrate the existence of an infinite class of perfect maps. Finally, we show by example that the existence of perfect maps is strongly dependent on the structure of the primary map as well as on the shape and structure of the submap.

II. Some Perfect Maps

A familiar example of a perfect one-dimensional map is a maximal length linear shift-register code.[1] For a shift register of length n, a maximal length shift register code is of length $2^n - 1$. Zierler has shown that all the $2^n - 1$ translates of an n-tuple $(a_0, a_1, \cdots, a_{n-1})$ of the code are distinct. Since $n = [\log (2^n - 1)]$, this code is an optimal $(2^n - 1) \times 1$ map on the torus. For example consider the binary difference equation

$$a_n = a_{n-2} + a_{n-3},$$

where $a_n = (0, 1)$ and addition is modulo two. Choose initially $a_0 = 0$, $a_1 = 0$, $a_2 = 1$, then the sequence

$$\{a_n\} = \{0\ 0\ 1\ 0\ 1\ 1\ 1\ 0\ 0\ 1\ 0\ 1\ 1\ 1\ \cdots\}$$

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[1] N. Zierler, "Linear recurring sequences," *SIAM*, vol. 7, pp. 31–48; 1959.

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is the solution of this equation. This sequence has period 7 and its set of three-bit translates,

$$\{(0\ 0\ 1), (0\ 1\ 0), (1\ 0\ 1), (0\ 1\ 1), (1\ 1\ 1), (1\ 1\ 0), (1\ 0\ 0)\},$$

are distinct.

dc Bruijn[2] (see also Golomb and Welch[3]) has shown that a maximum-length linear shift-register sequence can be increased from length $2^k - 1$ to 2^k by making the difference equation nonlinear. If the shift-register has length k and the difference equation for the sequence of period $2^k - 1$ is

$$a_n = \sum_{j=0}^{k} b_j a_{n-j},$$

then a sequence of period 2^k is generated by

$$a_n = \sum_{j=0}^{k} b_j a_{n-j} + a'_{n-1}a'_{n-2} \cdots a'_{n-k+1},$$

where the prime denotes complementation, *i.e.*,

$$a' = a + 1.$$

Since there are no periods of the sequence less than 2^k, the 2^k translates of length k must be distinct. Thus, a de Bruijn sequence of length 2^k is an example of a $2^k \times 1$ perfect map.

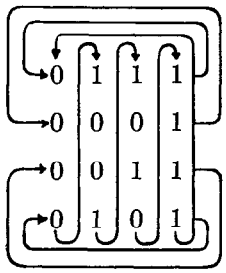
Set $k = 2m$ and in particular let $m = 2$. Then the difference equation

$$a_n = a_{n-3} + a_{n-4} + a'_{n-1}a'_{n-2}a'_{n-3}$$

generates the sequence

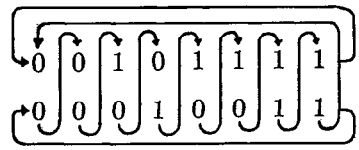
$$(0\ 0\ 0\ 0\ 1\ 0\ 0\ 1\ 1\ 0\ 1\ 0\ 1\ 1\ 1\ 1\ 0\ 0\ 0\ 0\ \cdots)$$

of period sixteen. Wrap the first sixteen bits of this sequence on a torus as follows:



Evidently this map is a perfect 4×4 map for a 1×4 submap. Similarly, one may construct a perfect $2^m \times 2^m$ map with $1 \times 2m$ submap by wrapping the first 2^{2m} bits of a de Bruijn sequence m times around a torus. Thus, we have established the existence of an infinite class of perfect maps.

In order to construct a perfect map with other than a linear or $(1 \times m)$ submap note that the above sixteen bit sequence can be wrapped in the following different manner about a torus:



Every 2×2 submap of this map is distinct. This map can be flattened into a planar 3×9 map by attaching a row and a column of boundary elements to the above matrix or to any of its translates. For example, if we attach a top row and a right column of boundary elements to the above matrix, we obtain

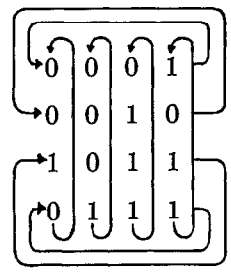
$$\begin{bmatrix} 1 & 0 & 0 & 0 & 1 & 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 1 & 1 & 0 \end{bmatrix}.$$

The sixteen 2×2 submaps of this matrix are different. Since there are two ways of attaching the boundary elements and eight translates of the toroidal map, there are 16 such perfect planar 3×9 maps derivable from the single 2×8 toroidal map.

We next investigate by example the possibility of a perfect *square* map for a *square* submap. Although we have been unable to generalize the example constructed, it is interesting for its own sake. It is the first example of a doubly-periodic binary function with no smaller periods.

III. Square Maps with Square Submaps

The matrix



and its transpose are the only perfect 4×4 maps on a torus for a 2×2 submap. The double periodicity of these two maps is apparent. To generate the above map consider a 2×2 submap:

$$A_{mn} = \begin{bmatrix} a_{m,n}, & a_{m,n+1} \\ a_{m+1,n}, & a_{m+1}, \quad a_{n+1} \end{bmatrix}.$$

The remainder of the map satisfies the two simultaneous matrix difference equations

$$A_{m+2,n} = A_{m,n} + \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

and

$$A_{m,n+2} = A_{m,n} + \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix},$$

[2] N. G. de Bruijn, "A combinatorial problem," *Koninkl. Ned. Akad. Wetenschap., Proc.*, vol. 49, pt. 2, pp. 758–764; 1946.

[3] S. W. Golomb and L. R. Welch, "Nonlinear Shift-Register Sequences," Jet Propulsion Lab., California Institute of Technology, Pasadena, Memo No. 20-149; October 25, 1957.

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where addition is componentwise and modulo two. Evidently this map is generated by an additive abelian group with four elements. The generators of this group are the matrices

$$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

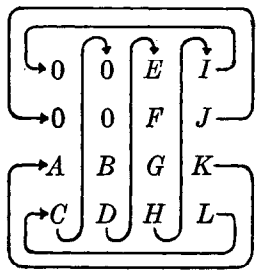
The transpose map is generated by interchanging the generators.

There is a total of 32 planar 5×5 maps obtainable from the above toroidal maps. For example, if we attach a left column and an upper row of boundary elements to the above matrix we have

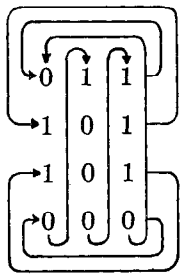
$$\begin{bmatrix} 1 & 0 & 1 & 1 & 1 \\ 1 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 & 1 \\ 1 & 0 & 1 & 1 & 1 \end{bmatrix}$$

as an illustration of a perfect 5×5 planar map for a 2×2 submap.

To show the strong dependence of a perfect map on structure the following matrix on a torus was investigated: This map differs from the previous map only in the boundary identifications. Here the sixteen elements are



helically placed on a torus. Since eight of the twelve letters in the matrix must be ones, there were $\binom{12}{4} = 295$ cases to consider. The 7090 IBM computer of the Rand Corporation was programmed by R. S. Grote to investigate this matrix for a 2×2 submap. It was found that there was no perfect map of this type for a 2×2 submap. An example of the closest approximation to a square map, connected in this manner, is given by the following nonperfect 3×4 map for a 2×2 submap:



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